Have you ever thought of observing the microorganisms present in rain water when you play? Have you ever wondered how milk turns into curd and which microorganisms are involved? It is clearly understood from previous unit that microorganisms can be seen only under microscopes. But microorganisms do not show much of its structural details under the light microscope due to lack of contrast and poor resolution. To improve the visibility of these tiny living organisms, stains and staining methods are of great use.

3.1 Techniques for Observing Microorganism

A considerable amount of information can be gained by careful microscopic examination of microorganisms. There are two general techniques used in the preparation of microbial specimens to observe them under microscope. First technique employs the unstained preparation of living cells and second one employs stained preparations of killed microorganisms.

3.1.1 Examination of Unstained Preparation

Living microorganisms can be examined directly by wet mount or by hanging drop preparations. Both the techniques are very useful in determining size, shape and motility of the microorganisms. The spirochetes (spiral bacteria) are normally examined in wet preparation through Dark-field microscope. Some cell inclusion bodies such as vacuoles and spores can be readily observed even without staining.

- A wet mount is made by keeping a drop of liquid containing microorganisms (culture) on a microscope slide and placing a cover slip over the drop. (Figure 3.1a)
- A hanging drop mount is made by using a cover slip and a cavity slide. Vaseline is applied on each of the four corner of the cover slip or around the cavity using a match stick. A drop of culture (liquid containing microorganisms) is placed on a cover slip. The cavity slide is placed upside down on the cover slip and inverted such that the drop is hanging (Figure 3.1b).

Since microbial cells are colourless and transparent, observation of microorganisms in wet preparation by bright field microscope is difficult. But, dark-field and phase contrast microscopes give contrast and make structures within the cells to appear clear. Therefore, these microscopes are useful for examination of unstained preparation.

3.1.2 Examination of Stained Preparation

Staining enables better visualization of microorganisms under a microscope. Microscopic examination of stained cells helps to reveal the size, shape and arrangement of microbial cells. Microbial cell staining is important in the identification of infectious pathogens.

3.2 Purpose of Staining

Staining is very useful for the following reasons:

- To make the microscopic semi transparent microbial cell visible.
- To reveal the size and shape of microorganisms.
- To demonstrate the presence of internal and external structures of microbial cells.
- To distinguish between different types of microorganisms.
- To produce specific chemical and physical reactions.
- To preserve the stained microorganisms as specimen slide.

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